

NEURAL BELLMAN OPERATORS: RESPONSE TO THE R8 ADVISORY REPORT

QIAN QI

Peking University

1. OVERVIEW

We thank the referee for the detailed examination of the R8 manuscript and its numerical archive. We have revised the original Neural Bellman Operators paper, retaining its economic topic, applications, prior results, and adverse evidence. The response is directed to the advisory report dated 3 October 2026 on candidate f9a7e32; it is not a response to the superseded R2 or R7 reports. The root article and supplement now integrate the new results rather than placing an additional protocol beside the paper.

The revision adds three substantive objects. First, an indexed critic guard enforces the selected-policy nodal target tolerance, with its memory and correction frequency recorded. Second, a signed action cover combines mean-value enclosures with verified face domination and a stopping-safe fallback. Third, the original dense continuous-time capital model receives a global optimal upper bound and an explicit feasible-policy lower bound. The numerical program recomputes complete-action certificates, retrains on genuinely nested grids, reports every historical full-domain refinement discrepancy, computes a defined compensating transfer, and records total costs. It retains the distinction between a mathematically valid envelope and a competitive neural proposal.

2. BLOCKING COMMENTS

B1. Identify the actor's contribution

We agree that the R8 comparisons did not demonstrate that the actor caused the successful local-search policies. The new abstract and conclusion state directly that the low-dimensional experiments do not establish an actor speed advantage. The title and topic remain Neural Bellman Operators: the object is now specified by

its proposal, guarded continuation, frozen deployment policy, and independent economic output. We do not rename local search as neural learning.

The new comparison holds the nine R8 hybrid policies fixed while replacing only their verifier. This isolates the contribution of the action-cover calculation. The fine-grid study additionally applies the identical critic guard to actor and actor-free search, and includes unguarded versions of both. Raw actors remain separately assessed. These are controlled statements about the implemented map, not a claim that the actor must dominate whenever the certificate improves.

B2. Reach useful complete-action accuracy and account for cost

The new signed cover uses validated partial-derivative intervals to bound payoff variation and reduce strict-sign coordinates to their maximizing faces. It does so only when every transition is wholly on one side of each stopping switch. Every other box retains the natural union enclosure. The theorem proves why each reduction preserves the supremum and why interruption preserves the final upper bound.

We rerun the shared complete-action envelope and independently propagate all frozen policy payoffs. The article prints each of the nine uniform nodal bounds and the number of one-step covers that close at the requested tolerance. The algorithm does not mark a budget-limited step as solved. A separate matched experiment compares the old and new cover on identical continuation vectors and feasible incumbents at three dates. It counts derivative work in elapsed time. Thus the cost comparison does not rely solely on dividing times from different machines or on treating a derivative query as free.

B3. Full-domain refinement and continuous-economy transfer

The earlier centre-only display was inadequate. The supplement now prints all eighteen historical full-domain maxima, upper quantiles, worst state-time locations, boundary-region maxima, and action saturation rates. In particular, the discrepancy exceeding two utility units near high risk aversion and the lower wealth boundary is retained and analyzed. No frozen coarse policy is replaced retroactively by a newly trained policy in that record.

We add fresh training on 17×25 and 33×49 grids, with twenty and forty dates. These levels are genuinely nested, unlike the full historical sequence. The guarded continuation is compared with the original fitted continuation under the same neural budgets and seeds. A separate complete-action envelope is computed on the fine grid and applied to all twelve fine-grid policies. This establishes a new nodal accuracy comparison rather than transporting a coarse-grid certificate by interpolation.

The guard supplies a proved evaluation perturbation of at most $\tau_h + \eta_h$, and its specified $0.1h^2$ tolerance addresses the evaluation component of a vanishing-perturbation argument. The comparison principle, transition consistency, and continuously monitored stopping errors remain separate requirements. We have not filled unknown continuous-state/time defects with zero or treated two grid levels as an asymptotic rate. The strengthened fine-grid calculation is therefore not described as a completed error bound for the stopped continuous diffusion.

B4. Strong baselines and the role of economic structure

The projection, classical recursion, direct-HJB, direct control optimization, and actor-free search results remain unchanged. We agree that exploiting low dimension is a legitimate advantage of those methods. The new evidence isolates a verified-enclosure improvement and a guarded-evaluation improvement; it does not use them to erase a classical comparator's better timing. Total costs include proposal construction, critic fitting, local search, reference construction, policy evaluation, independent verification, and, when reported, welfare conversion. Shared costs are amortized only across identical economies under stated numbers of policies.

B5. A specified map and substantive numerical consequences

The new article proves the guard's target-error bound, the signed cover's inclusion and face-domination properties, and a global bound for the original capital economy. The first gives an implemented output rule that meets a prescribed discrete evaluation tolerance; the second yields an independently measured tightening of the same policies' certificates; the third provides an absolute continuous-economy

benchmark. The proofs state their hypotheses and their costs. We do not claim a new Bellman principle, general convergence of constant-step Adam, or a universal worst-case complexity improvement from favorable derivative signs.

B6. Connect the high-dimensional evidence to an absolute benchmark

The dense nonlinear capital model is retained without modification. Jensen's inequality, bounded nonlinear drift, and an explicit common-consumption schedule yield an upper bound on its original continuous-time optimum and a feasible-policy lower bound on $\mathbb{R}^d$ from the stated initial condition. Outward time integration computes the bounds for dimensions ten and twenty. The schedule is also simulated on the exact Brownian paths used for the historical neural policies, and all paired comparisons are retained.

These bounds are deliberately identified as global model benchmarks, not tight loss certificates for each neural policy. They do not convert a sampled residual into a uniform one, or subtract an unknown Euler bias from a continuous-time bound. They nevertheless remedy the absence of any decision-relevant absolute performance benchmark in the former paired-policy study. A simultaneously high-dimensional, tight, fully continuous neural-policy certificate is not asserted as an accomplished result.

B7. A coherent argument without discarding the economic scope

The original recursive-utility, endogenous-preference, temporal-self, dynamic-game, nonsmooth, and diffusion-trace applications remain. The main argument is reorganized around a defined policy map and three error decisions: evaluation output, action maximization, and economic transfer. The new result section connects the guard, signed cover, nested-grid experiment, and welfare interpretation in the same preference economy, then gives the absolute benchmark in the existing capital economy. Earlier evidence remains identifiable as history. Every inherited main-text label and section is preserved, and the complete pre-revision manuscripts are archived.

3. MAJOR COMMENTS

M1. Title and abstract

The title remains Neural Bellman Operators, consistent with the original topic. The abstract now identifies the guarded map, signed enclosure, and absolute capital benchmark, and states that the low-dimensional evidence does not establish an actor speed advantage. It does not imply that a proposal-agnostic certificate proves the superiority of its neural input.

M2. Full-domain statistics

All historical refinement configurations now have maxima, 95th and 99th positive-advantage quantiles, worst locations, boundary summaries, and action saturation rates in printed tables. The machine-readable record additionally retains all specified quantiles and the fraction of nodes above 0.1. The centre is no longer the only printed location.

M3. Economic meaning of utility accuracy

We define an externally financed flow-consumption supplement, with the policy, state dynamics, stopping bequest, and adjustment costs unchanged. Its uniformly sufficient percentage is verified against the optimal upper envelope over all active starting nodes and dates. This is not a self-financed feasible-policy comparison or a common homothetic consumption equivalent. Large required supplements are reported as economically substantial rather than being described as a small loss because an absolute utility threshold passes.

M4. Total cost at achieved accuracy

The article contains a matched one-step verifier comparison and the supplement an explicit total-cost account for one, nine, and one hundred policies. The latter labels historical R8 training times separately from new verification times; it is not represented as a same-machine training race. The fine-grid comparison records fresh training, separate verification, and correction storage. Welfare conversion has its own recorded cost.

M5. The action box

The box is treated as the primitive feasible set of the constrained economy, not a computational approximation to an unspecified unconstrained optimum. The action-bound sensitivities remain and the new saturation tables show binding margins. No economic interpretation is carried beyond the declared consumption, leverage, and adjustment constraints.

M6. Deployment

The certified preference policy is an explicitly stored state-time action table, constructed from the actor or actor-free proposal and local improvement, with its guarded continuation when used. Node lookup entails no additional optimizing queries. Raw network deployment and off-grid interpolation are separate objects; the latter retains the documented failure study. Every indexed correction and its storage cost is saved.

M7. Dynamic games

The complete dynamic best-response experiments remain finite state/action game results. No continuous-action or continuous-capital error is inferred. Their role is a strategic validation laboratory within the original NBO scope, with complete unilateral deviations rather than one-step gradient tests. Existing game arrays and adverse fitted-stage results are not deleted.

M8. A high-dimensional economic benchmark

The new feasible schedule and global relaxation bound apply to the original non-linear capital economy. Their derivation and interval computation are explicit, and the schedule provides a paired policy benchmark under the saved Brownian increments. The bound is not presented as tight when it is not, and simulation confidence intervals remain limited to Monte Carlo error at a stated Euler discretization.

M9. Interval terminology

The article, tables, and response uniformly qualify interval results by the inherited binary64 execution contract. Polynomial transcendental enclosures and square-root endpoint checks support the numerical inclusion arguments. Regression stress tests do not formally verify the interpreter, processor, or library implementation.

M10. Scholarly architecture and historical preservation

The introduction, abstract, conclusion, and current result chain are rewritten around the numerical map rather than the order of review exchanges. The old manuscripts and evidence remain in the archive, and inherited sections are retained to preserve the original economic research scope. The response history is not used as proof of a result; each current table is generated from named raw evidence and an exact execution identity.

4. REPRODUCTION AND REMAINING SCOPE

The new branch contains code, proofs, raw policies, neural weights, indexed corrections, complete-action arrays, seed histories, failure and budget flags, full-domain tables, source hashes, compilation records, and this response. The workflow reruns the new numerical program and inherited tests, compiles the Econometrica-class article and supplement, and commits the authoritative evidence on a new branch. A test success records software and accounting checks, not journal acceptance or the success of every economic threshold.

The unresolved distinctions are explicit: the stopped preference economy still lacks a numerical bound for every continuous-state/time and monitoring defect; a low-dimensional actor advantage is not established; and the global capital bracket is not a tight continuous neural-policy regret bound. These are not replaced by a change in topic or by an assertion that the original research program should be abandoned. The revision adds verified computations and theorems within that program and makes their exact implications available for further review.